

Midterm 2 for COSC330/PHYS306 - Fall 2021

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1 Chapter 6 - Functions of Combinational Logic

1. Consider Fig. 1, in which a 4-bit adder is depicted. Assuming a uniform worst-case delay of 8 ns from carry-in to carry-out for each stage, the total delay is 32 ns. (a) At what maximum frequency can this circuit perform additions, if the delay must be less than a clock period? (b) What would the result in (a) be if the adder was extended to 8 bits? (c) Create a timing diagram showing the addition of the numbers in Fig. 1. **Bonus:** include the timing delays, and indicate the clock period.

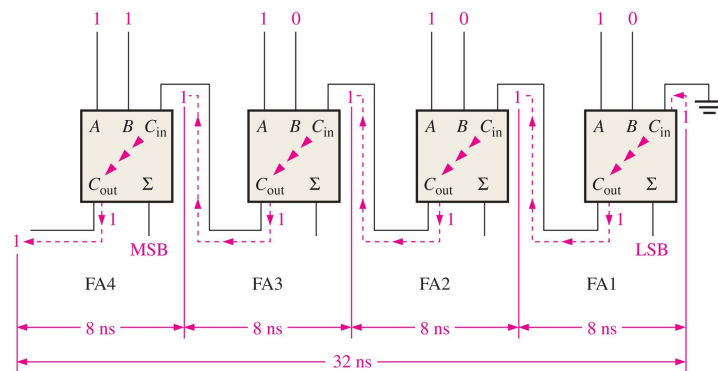


Figure 1: A schematic of a 4-bit adder with delays marked in nanoseconds.

2. Using AND, OR, XNOR, and inverter gates, (a) create a 2-bit comparator that yields True if $A > B$, (b) True if $A = B$, and (c) True if $A < B$. For each circuit, assume both A and B are 2-bit binary positive numbers. (d) Wrap this all into one circuit with three outputs and 4 inputs. *Hint: use Karnaugh maps for parts (a)-(c) to simplify the gates.*
3. Generate the logic gates that convert 4-bit gray code to 4-bit binary code, and show that it works with a timing diagram.
4. Consider Fig. 2, in which a decimal to binary conversion circuit is shown. (a) Add logic to the circuit such that it becomes a hexadecimal to binary converter. (b) Draw a logic symbol for this circuit with the correct number

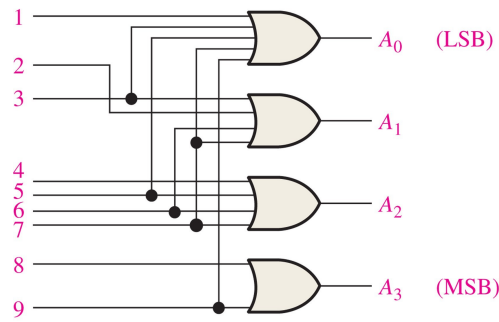


Figure 2: A decimal to binary encoder circuit.

of inputs and outputs. (c) Connect two hex-to-bin converters to a symbolic 2-to-1 multiplexer with the correct number of data select line(s) to form a system that can send two-digit hex numbers over one output line.

2 Chapter 7 - Latches, Flip-flops, and Timers

- Consider Fig. 3, in which a divide-by-four frequency divider is depicted with D flip-flops and a CLK signal. (a) Elaborate on this circuit to create a circuit that can divide the clock frequency by 2, 4, or 8. Show the flip-flops explicitly. (b) Develop a symbol for the 2-4-8-16 divider, with CLK signal input and four outputs (one each for $f/2$, $f/4$, $f/8$, and $f/8$, where f is the clock frequency). (c) Connect the symbol for the new part to a symbolic 4-to-1 multiplexer with the appropriate number of data select lines to form a clock division system. (d) Show with a timing diagram the output if the user changes the data select lines at least once. For parts (b)-(d), it is only necessary to show symbols rather than individual flip-flops.

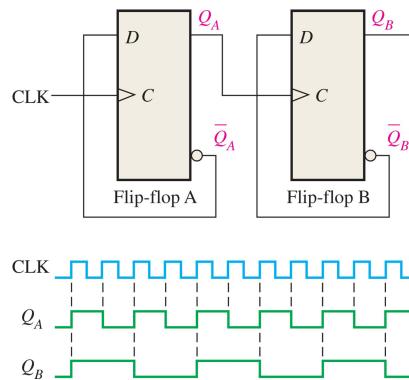


Figure 3: Two D flip-flops forming a frequency divider.

3 Chapters 8 and 9 - Shift Registers and Counters

1. Consider the timing diagram in Fig. 4. Using any combination of *shift registers* and supporting gates, create a circuit with 8-outputs that produces this timing diagram.

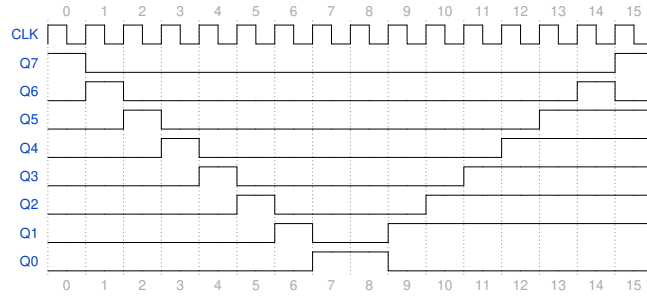


Figure 4: A waveform that repeats every 16 clock periods.

2. Consider Fig. 5, in which 4 and 5-bit Johnson counters are depicted. (a) Create a circuit based on Fig. 5 called a *combinatoric trigger*. Imagine four digital channels A, B, C, and D, as inputs. The output of the circuit should be True if *any two* of the four channels is True *within 100 ns of each other*. Develop any necessary logic symbols, and use the appropriate clock frequency. The Johnson counter(s) can have any number of bits.

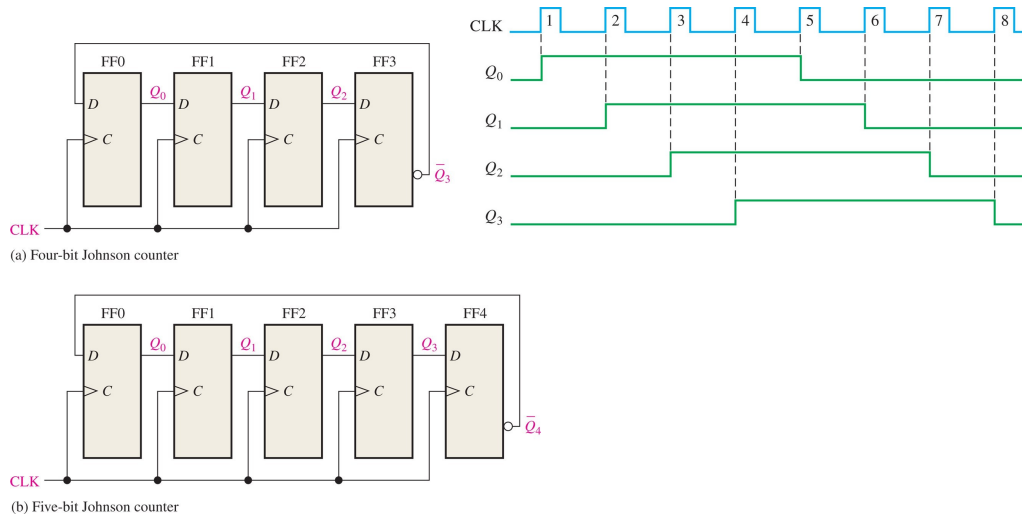


Figure 5: (Left) 4 and 5-bit Johnson counters. (Right) The timing diagram for the 4-bit case.

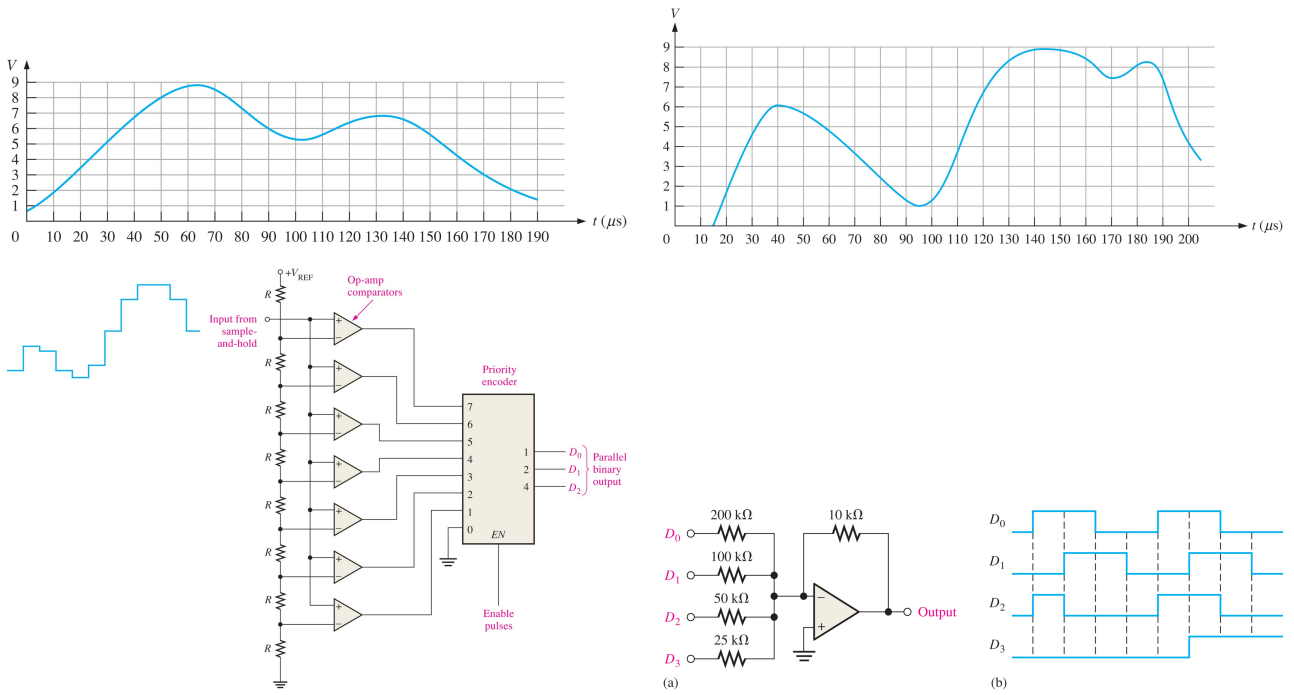


Figure 6: (Top left) An input waveform for a 3-bit flash ADC with $V_{CC} = 8$ Volts. (Top right) A second input waveform for a 3-bit flash ADC with $V_{ref} = 8$ Volts. (Bottom left) The 3-bit flash ADC. (Bottom right) A 4-bit DAC with binary weighted inputs.

4 Chapter 12 - Signal Conversion and Processing

- Consider Fig. 6. (a) Determine the binary output codes from the 3-bit flash ADC in Fig. 6 (bottom left), given the input waveform in Fig. 6 (top left). (b) Repeat (a), but for the input waveform in Fig. 6 (top right).
- Determine the output of the DAC in Fig. 6 (bottom right), if the sequence of 4-bit numbers is applied to the inputs. The data inputs have a low value of 0V and a high value of 5.0V.